Taichi Kamei

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Education

3rd Year Engineering Physics *Expected Graduation - May 2028*
University of British Columbia | Vancouver

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Thermodynamics, Statistical Mechanics, Quantum Mechanics

Dean's Honor List 2025

Experience

Quantum Photonic Researcher *May 2026 - Present*
UBC Blusson Quantum Matter Institute

Projects

Drone Flight Controller *Feb 2026 - Present*

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Sized charge and discharge FETs for the BMS IC based on anticipated continuous/pulse drain current, DS voltage, and on-resistance to achieve safe handling and minimize heat dissipation
- Designed a Sallen-Key low-pass filter with non-inverting amplifier for amplifying 5kHz signal while attenuating 25kHz PWM noise, simulated its functionality on NgSpice
- Designed a custom 4-layer GPS & magnetometer PCB on KiCAD fitting within 260mm by 260mm
- Prototyping a flight controller PCB, integrating ESP32-S3-Mini-1, BMS, IMU, and a barometer

Autonomous Clue Detecting Robot *November 2025*

- Developed an autonomous vehicle in Gazebo and ROS 1 using OpenCV for PID control driving, obstruction avoidance, and CNN clue detection
- Designed and implemented a finite state machine (FSM) capable of transitioning its states for moving obstruction detection, clue board detection, and drive under various track surface conditions
- Optimized a PID algorithm for precise center-lane driving using dual side lines instead of a typical single line-following, overcoming camera center offset from the geometric center of the lane

Autonomous Payload Retrieving Robot *June - August 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

Skills

Software	Python, Java, C/C++, Assembly, VHDL, Linux, Git
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, DMM, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper
Embedded	ESP-IDF, Arduino, Raspberry Pi, FPGA, ROS, I2C, SPI, UART